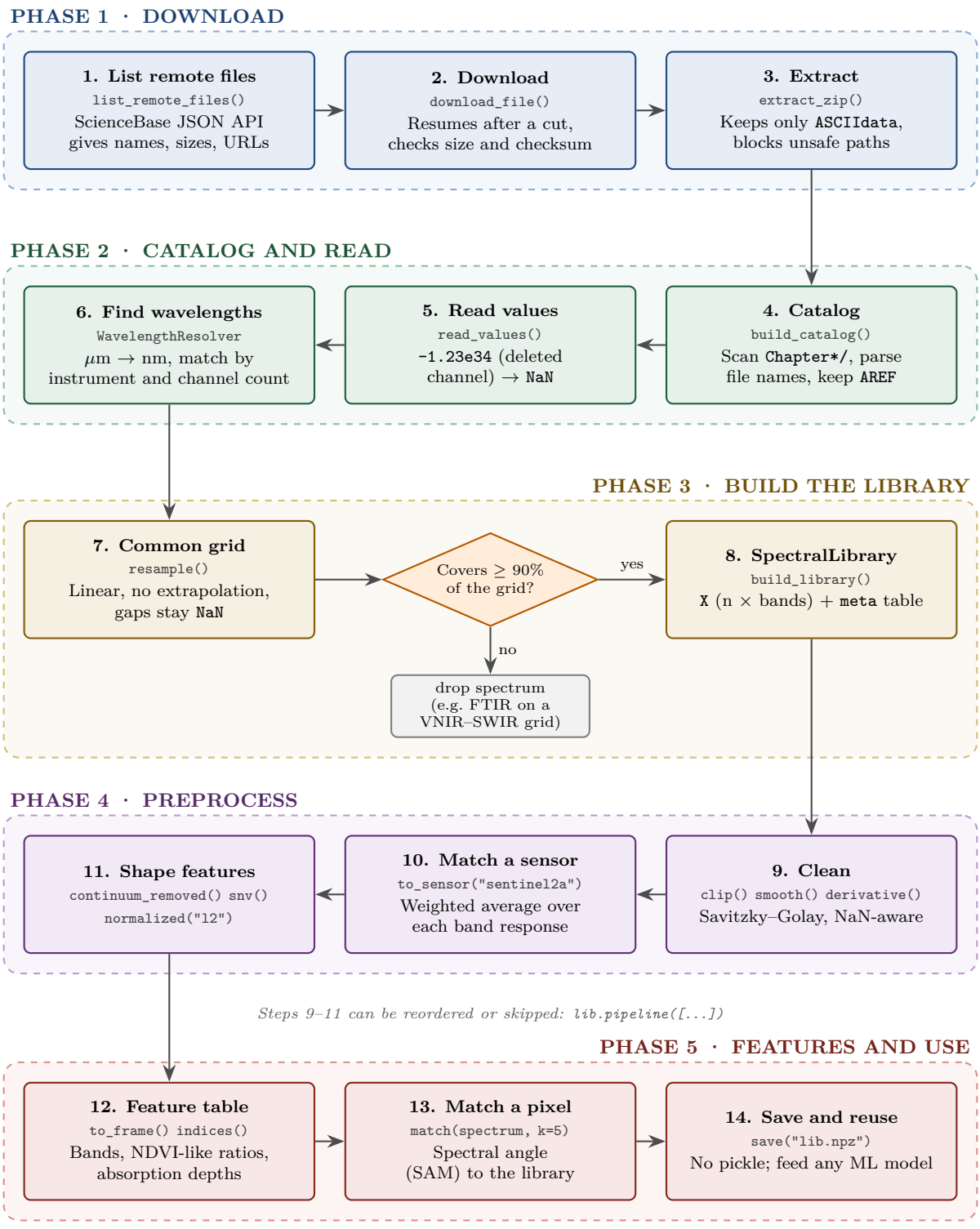
usgs-splib07: from download to features

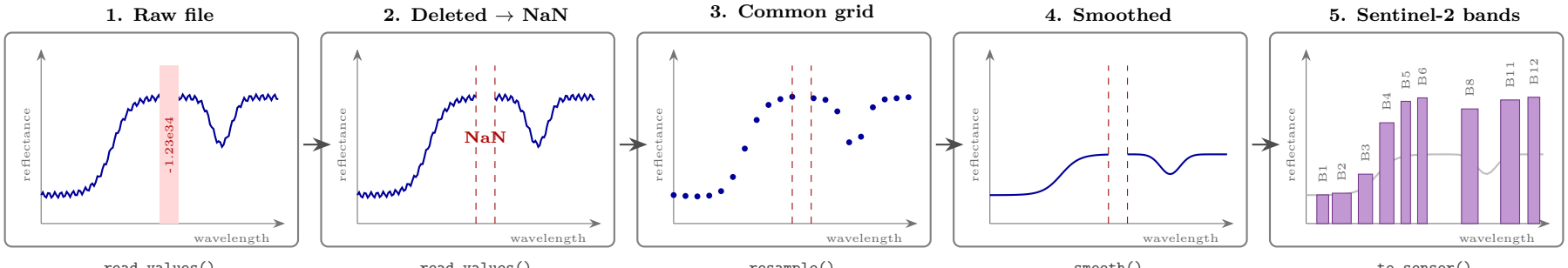
Read it top to bottom; each row is one phase, arrows follow the data.



Data: Kokaly et al. (2017), USGS Spectral Library Version 7, doi:10.3133/ds1035

The journey of one spectrum

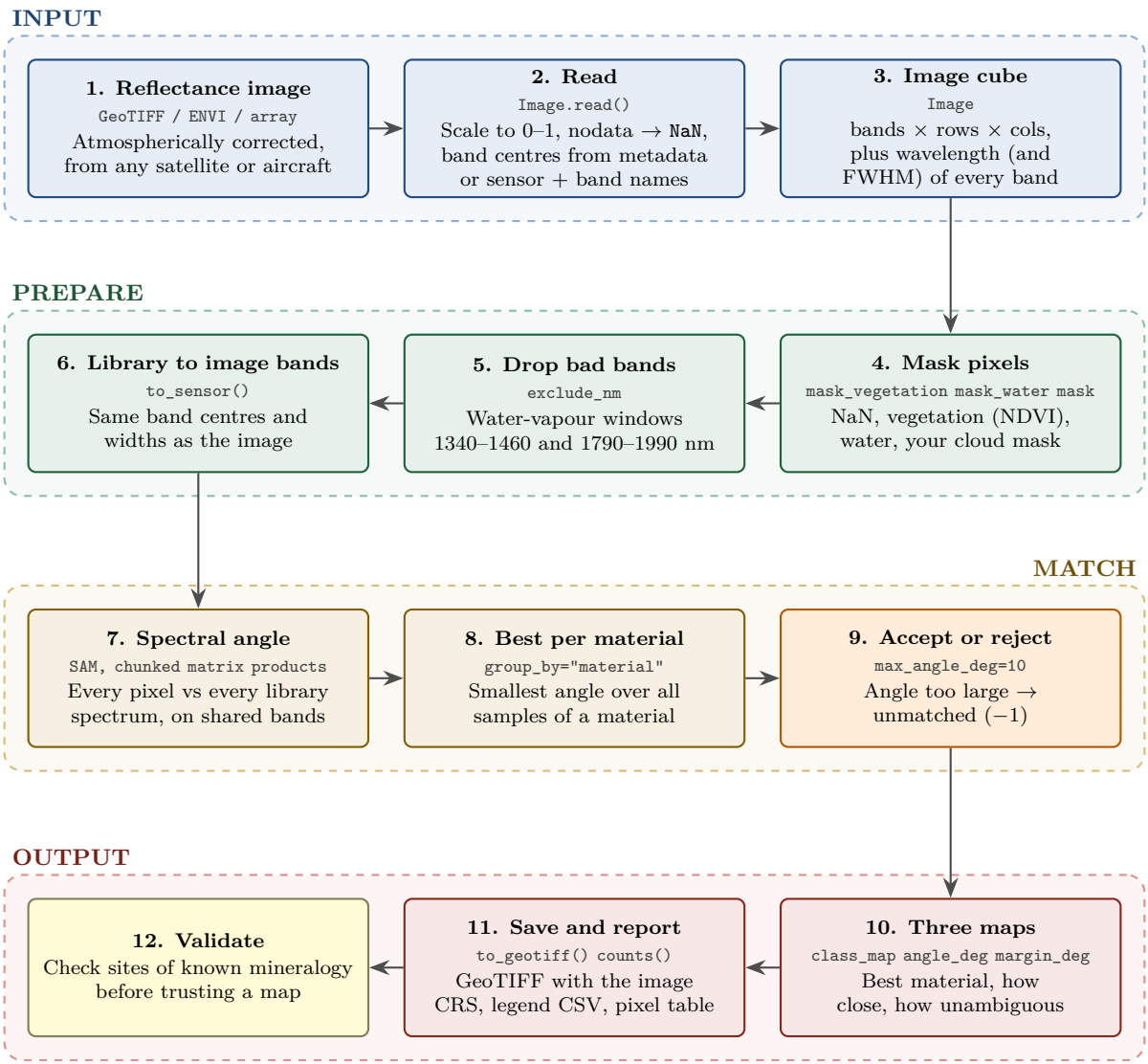
Same curve at every stage: watch what each function changes.



Why it matters: a gap that is silently interpolated would invent reflectance values in the water-absorption band. Keeping it as `NaN` lets later steps (coverage test, band convolution, spectral angle) ignore those channels honestly.

Matching a satellite image to the library

Same steps for Sentinel-2, ASTER, or a 200-band imaging spectrometer.



More narrow bands separate more minerals: a 13-band sensor blends the 2165/2205 nm clay doublet, a 10 nm spectrometer resolves it.